

About this app

How the Merchant app fits into the platform.

A separate app, not a separate service

`/merchant` is its own React/Vite app (port `5175` , alongside consumer `5173` and admin `5174`) — but it talks to the **same Express backend on port 3001**. There is no separate merchant service or database; route files are mounted on the existing server (`memberAuth.routes.ts` , `settlement.routes.ts` , `merchantSelf.routes.ts`).

Why a separate app

A merchant is a business with staff turnover and shared tills — a different trust context from an individual consumer. The merchant app has **no wallet, no wagmi, no RainbowKit**. Custody (one corporate account) is split from access (many operators with passkey login and roles) and execution authority (settlement above a threshold needs second approval).

Org / member model

A merchant can have many **operators** (`merchant_members`), each with a role:

Role	Can do
cashier	Ring up Point of Sale, view Sales and Products (read-only)
store_manager	Same as cashier (reserved for future store-scoped limits)
org_admin (head office)	Team, My Business, Products catalog, settlement approval above threshold

Operators sign in with a **passkey** (WebAuthn) — not a Safe wallet signer. First login is a claim ceremony via invite ID + email.

Custody — why no new wallet was needed

Receiving needs no signature (`trustedCounterparty` push). `withdrawToExternal` is `onlyRole(ADMIN_EXECUTOR_ROLE)` — backend-signed, no merchant private key. Head-office settlement approval is an **off-chain** rule before that call.

Settlement threshold & approval

Any operator can request a payout. At/below threshold it executes immediately; above threshold a **different** `org_admin` must approve. Settlement resolves fiat currency to pegged on-chain token (e.g.

ZAR → TTZA).

Origin note for WebAuthn

`WEBAUTHN_ORIGIN` is a comma-separated list so consumer (5173) and merchant (5175) can complete passkey ceremonies against the same backend.

API — member auth (no wallet)

Method · Path	Auth	Purpose
POST /api/member-auth/invite	Org admin	Invite operator → memberId
POST /api/member-auth/claim-options	Public	WebAuthn registration challenge
POST /api/member-auth/claim	Public	Claim invited seat with passkey
POST /api/member-auth/login-options	Public	WebAuthn login challenge
POST /api/member-auth/login	Public	Verify assertion → member JWT
GET /api/member-auth/me	Member JWT	Current operator + org
GET /api/member-auth/members	Org admin	Team roster

API — settlement

Method · Path	Auth	Purpose
GET·POST /api/settlement/config	Member / Org admin	Payout threshold
GET·POST /api/settlement/requests	Member	List / request payout
POST /api/settlement/requests/:id/approve	Org admin	Approve pending request
POST /api/settlement/requests/:id/reject	Org admin	Reject pending request

Change vouchers (digital change at till)

When a customer would receive coins as change, a cashier can issue a **change voucher** — credit to their wallet. Add a product with delivery type **Voucher**.

Step	Who	What happens
1	Cashier (POS)	Issue change voucher → QR or Send to @tag

Step	Who	What happens
2a	Customer (QR)	Receive → scan change voucher QR
2b	Customer (link)	Open deep link in Receive
2c	Customer (@tag)	Direct credit to @tag
3	On-chain	Debit merchant claim, credit consumer (15 min QR expiry)

API — merchant self-service

Mounted at `/api/merchant` — resolves merchant from member JWT. Powers POS, Sales, Products, My Business.

Method · Path	Auth	Purpose
GET <code>/api/merchant/me</code>	Member	Own profile
PATCH <code>/api/merchant/me</code>	Org admin	Edit business details
GET·POST <code>/api/merchant/me/products</code>	Member / Org admin	POS catalog
GET·POST <code>/api/merchant/me/stores</code>	Member / Org admin	Store locations
GET <code>/api/merchant/me/sales</code>	Member	Sales ledger
POST <code>/api/merchant/me/change-voucher/*</code>	Member	Issue change vouchers

Not yet built

Invite-link/email-token flow; bank off-ramp for settlement (destination must be on-chain address today); per-store product catalogs (products filtered by currency at POS today).